Module 1: Introduction to Convolutional Neural Networks (CNN)

1. What is CNN?

A Convolutional Neural Network (CNN) is a type of Artificial Neural Network mainly used to process images and identify visual patterns.

CNNs can learn features such as edges, lines, textures, shapes, and object parts directly from image data.

Applications of CNN include:

- Image classification
- Face recognition
- Object detection
- Medical image analysis
- Self-driving cars
- Handwriting recognition

2. Why do we need CNN?

An image is made up of pixels. Each pixel contains a numerical value representing its intensity or colour.

For example, a 128 × 128 RGB image contains:

128 × 128 × 3 = 49,152 pixel values.

A basic fully connected ANN can convert this image into a one-dimensional vector. However, it may require a huge number of parameters because every input can connect to many neurons.

Flattening also removes the explicit two-dimensional arrangement of the pixels.

CNN solves this problem by:

1. Using small filters to examine local regions.

2. Reusing the same filter across the image.
3. Learning useful features automatically.
4. Reducing spatial dimensions when required.

3. How does CNN understand an image?

Imagine a CNN trained to classify cats and dogs.

- Early layers may learn edges and lines.
- Middle layers may learn textures, curves, and shapes.
- Deeper layers may learn combinations of patterns, such as parts of animals.
- The final classification layers use these features to predict the class.

These features are learned during training.

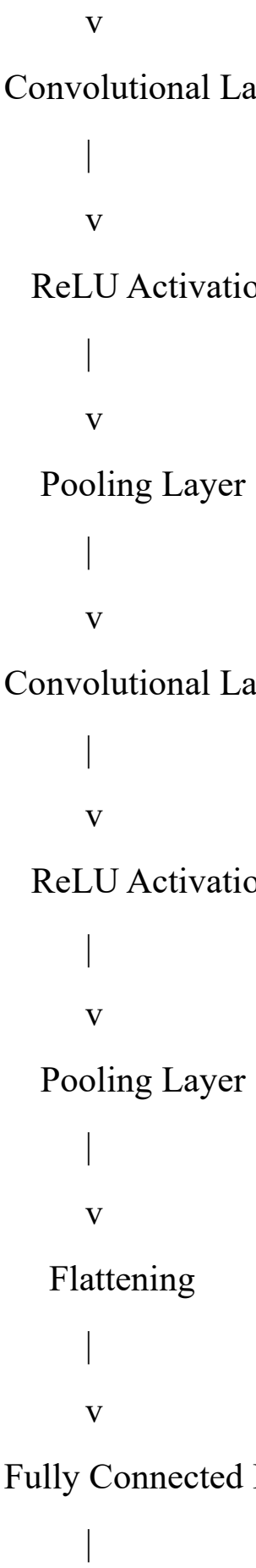
4. Main components of CNN

1. Input image
2. Convolutional layer
3. Activation function, such as ReLU
4. Pooling or downsampling
5. Additional convolutional layers
6. Flattening or another feature aggregation method
7. Fully connected layer
8. Output layer

5. CNN architecture diagram

Input Image

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Convolutional Layer
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v
ReLU Activation
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v
Pooling Layer
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v
Convolutional Layer
|
v
ReLU Activation
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v
Pooling Layer
|
v
Flattening
|
v
Fully Connected Layer
|

v

Output Class

This is a simplified classification architecture. Real CNNs may use different structures.

6. Important distinction

Image classification predicts what an image contains.

Object detection predicts what objects are present and where they are located, commonly using bounding boxes.

Image segmentation predicts a class for individual pixels or regions.

Conclusion

CNNs are powerful for image processing because they learn local visual features using filters and reuse their weights across an image. This often makes them more efficient than a basic fully connected ANN for image tasks.